Don't Blink: Evidence Collapse during Multimodal Reasoning

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WORKSHOP PAPER
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MAIN CLAIM

Visual grounding decays during reasoning, but disengagement warrants action only when the task still needs the image.

Text uncertainty transfers best globally; the same vision check can help on visual-reference tasks and hurt on symbolic tasks.

9/9 model × dataset cells show visual-attention decay	**2,365** deduplicated samples in interaction analysis	**0.723** cross-dataset AUROC for full-response entropy	**18** directed cross-dataset transfers test actionability

1 The interpretability gap

● FAILURE EXEMPLAR · HALLUSIONBENCH

A confident answer can lose its image

87.2% entropy-derived confidence on a wrong map answer

Looks at the map → Textual prior takes over → Wrong answer, no grounding recovery

Here GLM-4.6V-Flash answers from a prior about state size, although the supplied map supports the opposite answer. Text confidence alone cannot expose this drift.

● WHY TRAJECTORIES?

Reasoning can outlast grounding

Image tokens stay fixed while generated text keeps growing. The model may gradually rely on its own textual prior instead of the pixels that should support the answer.

The operational question is whether image use ended too early or the task had already moved to valid symbolic reasoning.

● MEASUREMENT

Track evidence use across the trace

Image + question (visual tokens and generated reasoning) → **Grounding layers** (six layers chosen by held-out evidence localization) → **Seven checkpoints** (five thinking, two answer positions)

At each checkpoint, **V** measures attention mass on all visual tokens. $\mathbf{A_{bbox}}$ restricts it to annotated evidence regions.

$$V_{\text{thinking AUC}} = \text{area under the five thinking-phase V values}$$

This trajectory measure distinguishes sustained engagement from early collapse, even when both runs end with similar answer-time attention.

Grounding layers differ by architecture

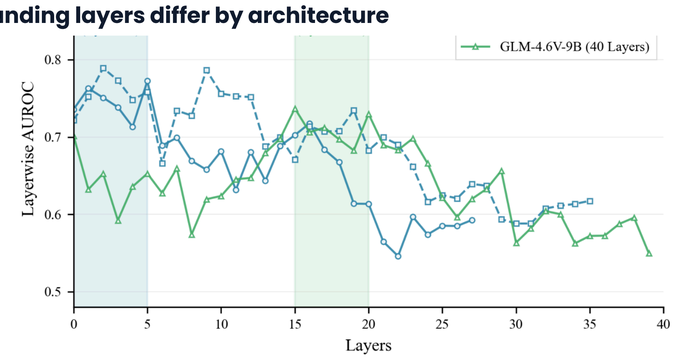


Fig. 1. Qwen3-VL-2B and 8B peak near layers 0–5; GLM-4.6V-Flash peaks near layers 15–20. Layers were selected on 100 held-out Visual-CoT samples.

● EVALUATION SCOPE

Three models × three datasets

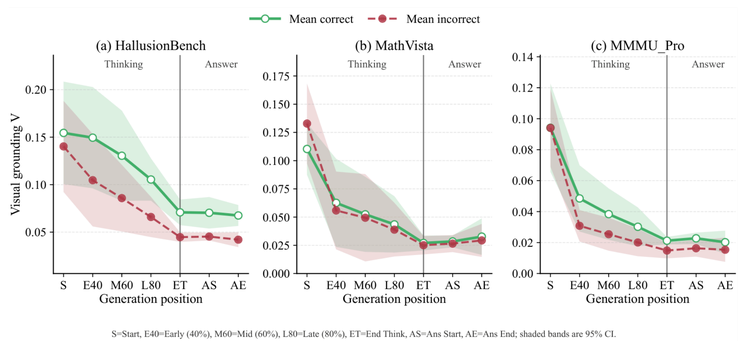
Qwen3-VL-2B · Qwen3-VL-8B · GLM-4.6V-Flash

HallusionBench · MathVista · MMMU Pro

Cross-dataset testing uses six directed transfers per model, for 18 total transfers.

2 Evidence collapse is measurable

Visual attention falls through thinking and answer



S=Start, E40=Early (40%), M60=Mid (60%), L80=Late (80%), ET=End Think, AS=Ans Start, AE=Ans End; shaded bands are 95% CI.

Fig. 2. Mean visual attention at seven positions; shaded bands are 95% confidence intervals. Correctness separation peaks during reasoning in 7/9 cells; MathVista shows a crossover.

Why an endpoint is insufficient.

Reasoning-time attention often separates correct from incorrect answers more strongly than answer-time attention. A final-token check can miss the trajectory that led to the answer.

● TRAJECTORY SHAPES

Different errors leave different traces

Persistent deficit: incorrect HallusionBench traces, and GLM on MMMU Pro, stay less grounded.

Crossover: incorrect MathVista traces can start more engaged, then fall below correct traces.

Moderate separation: Qwen on MMMU Pro has smaller, less stable gaps.

● EVIDENCE-REGION ATTENTION

The decline reaches relevant evidence

94.3–100% of samples per cell lose total visual attention

83.5–100% of samples per cell lose attention on evidence regions

Both total visual attention and evidence-region attention decline in every model × dataset cell. Relative evidence-region loss ranges from **53.3% to 90.8%**.

Attention is correlational; the observed decay alone does not establish that it caused an error.

● UNCERTAINTY MONITORS

Entropy is the best global monitor tested

Cross-dataset probe	AUROC ↑	Brier ↓	ECE ↓
Full-response entropy	**0.723**	**0.147**	**0.075**
Entropy + vision	0.695	0.155	0.082
Thinking visual AUC only	0.576	0.165	0.083



Incorrect answers have higher full-response entropy in all nine cells; this signal exceeds answer-span entropy in 8/9 cells. Global vision fusion worsens these mean metrics.

3 When should grounding trigger action?

"Confident but blind" changes meaning by task

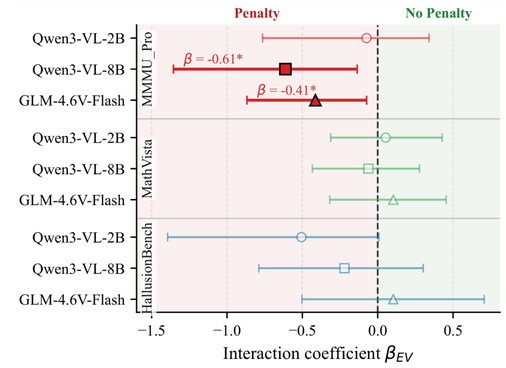


Fig. 3. Negative β_{EV} means low entropy plus low visual engagement predicts extra error. The penalty is significant for Qwen3-VL-8B and GLM on MMMU Pro; MathVista has no such penalty.

● INTERACTION MODEL

Test whether blindness changes confidence

$$\text{logit } P(\text{correct}) = \beta_0 + \beta_E E + \beta_V V + \beta_{EV} EV + \beta_L L$$

E is full-response entropy, V is thinking-phase visual engagement, and L controls for response length; each is standardized. A negative β_{EV} marks an extra confident-but-blind penalty.

● VISION-VETO STRESS TEST · 90% COVERAGE

The same veto can help or hurt

Defer by full-response entropy → Veto bottom 5% by visual engagement → Restore 90% coverage

Transfer target	Model	Δ risk (pp)
MMMU Pro	GLM	**−1.94**
MMMU Pro	Qwen-2B	**−0.66**
HallusionBench	GLM	**−1.11**
MathVista	GLM	**+1.87**
MathVista	Qwen-2B	**+1.86**
MathVista	Qwen-8B	**+1.20**

Negative Δ risk means fewer errors among accepted answers. Across all 18 transfers, the veto helps 6 and hurts 12; its direction matches the interaction sign in every transfer.

● ACTIONABLE INTERPRETATION

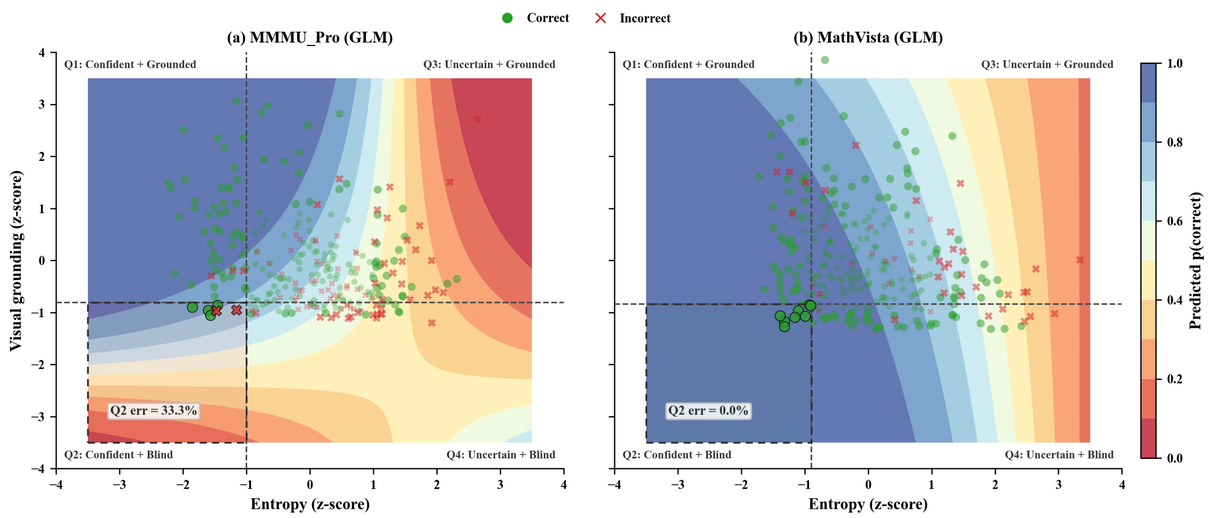
Route the check by task structure

Keep consulting the image?
Sustained visual-reference tasks: visual disengagement can flag a confident error. A conditional vision check may improve deferral.

Extract, then reason?
Symbolic tasks: early extraction can make later disengagement benign. A global veto can reject good answers.

This is a diagnostic rule supported by the measured task regimes, not a deployed policy. Task characterization is needed before using a vision veto.

4 One signal, two task regimes



● SAME SIGNAL · OPPOSITE DECISION

MMMU Pro — **33.3%**

GLM's low-entropy, low-grounding corner has this observed error rate. Its interaction penalty is $\beta_{EV} = -0.414$, and the vision veto reduces selective risk by 1.94 points.

MathVista — **0.0%**

The analogous GLM corner has this observed error rate. Its interaction is weakly positive and the same veto raises selective risk by 1.87 points.

Decision rule suggested by the evidence

Start with full-response entropy. Use visual disengagement to question an answer only when the task demands sustained visual reference.

Fig. 4. GLM regime maps show model-predicted correctness and observed samples. Extreme-corner counts are small; the continuous interaction is the primary evidence. Attention is correlational; results cover three models and three datasets.

TAKEAWAY

Use full-response entropy as the default uncertainty monitor. Add grounding-based intervention only where continued visual reference is required.

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